TRIX: Training-Free Reranking for Instruction-Following Retrieval with Exclusions

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Abstract

Retrieval requests often combine a target topic with positive criteria, background context, and explicit exclusions. Frozen retrievers rely heavily on lexical and semantic entity matching, overlooking how these instructions define relevance. This failure is acute for exclusions, where an excluded entity can become a strong matching signal. Existing remedies often require instruction training or representation adaptation. We introduce TRIX, a training-free reranking framework for frozen retrievers. TRIX preserves the complete request for retrieval and decomposes it into affirmative and exclusion views, focusing positive evidence and removing irrelevant background context. The two views can still share topic information, causing raw exclusion scores to penalize relevant content. TRIX removes the exclusion similarity predicted by affirmative relevance, then uses the residual to reduce positive scores and suppress exclusion matches. It requires no additional training data, model updates, or document re-encoding. Across three benchmarks and four frozen retrievers, TRIX improves retrieval effectiveness and instruction sensitivity, increasing the latter by up to 16.4 points and outperforming the strongest reported training-free baseline on retrieval with exclusion constraints by up to 3.3 points.

1 Introduction

Retrieval requests often combine a target topic with positive relevance criteria, background context, and explicit exclusions. Frozen retrievers rely mainly on lexical and semantic matching, but these signals do not reliably capture how each part of an instruction affects relevance. The problem is especially clear for exclusions because the excluded entity remains part of the request. In lexical retrieval, it contributes matching terms like any other query content. Dense retrievers face a related limitation: pooled embeddings compress topical content and compositional roles into a single vector, and cosine scoring can keep texts with similar content but different negation or scope close in the embedding space (Ralev et al. 2026). A document can therefore receive a higher score for matching an entity that the user explicitly asks to exclude. Figure 1 illustrates a request for evidence that a US college education improves employment or income while excluding documents about Manhattan colleges such as Columbia University. Documents about employment outcomes for Columbia graduates match the topic and entities in the request, yet they violate the exclusion. This behavior agrees with evidence that neural rankers are often insensitive to meaning changes introduced by negation (Weller, Lawrie, and Durme 2024; Weller et al. 2025a).

Separating the instruction into positive and exclusion views is a natural way to make these roles explicit. However, the two views can still share topic information. In Figure 1, the fifth-ranked document satisfies the request and does not discuss the excluded Manhattan colleges, but retrieval scores are based mainly on lexical or embedding similarity. Its broader discussion of US colleges and graduate outcomes therefore raises its scores under both the affirmative and exclusion views. Directly subtracting the exclusion score would penalize a relevant document. An effective correction must distinguish evidence of the excluded content from similarity caused by the shared topic.

Prior methods introduce instruction control through instruction-trained retrieval, inference-time query adaptation, or logic-based constraint modeling. Instruction-trained retrievers such as Promptriever (Weller et al. 2025b), Dual-View Training (Zeng et al. 2026), and InstEmb (Gao et al. 2026) learn instruction-conditioned representations, but require additional supervision and model training. Inference-time query adaptation methods such as DEO (Lee et al. 2026) modify the query embedding at inference time, but direct optimization can move the query vector away from the embedding structure learned by the retriever, making its similarity to document embeddings less reliable. Logic-based constraint methods such as NS-IR (Xu et al. 2025) and OrLog (Hoveyda et al. 2026) translate instructions into logical structures and apply symbolic or probabilistic inference over the retrieved candidates. The additional translation, candidate-level model calls, and formal inference increase online retrieval latency.

We therefore introduce TRIX, a training-free reranking framework that corrects the scores of a frozen retriever without updating its representations, re-encoding documents, or adding candidate-level constraint inference. TRIX retains the complete instruction for initial retrieval, derives an affirmative view that focuses the positive criteria and removes irrelevant background context, and represents excluded content in a separate view. It uses a query-local residual as a proxy for exclusion evidence beyond topical match. Based on this signal, the final scoring rule preserves or reduces affirmative support and applies an exclusion penalty.

Across an instruction-following benchmark (FollowIR) and two exclusion-aware benchmarks (NegConstraint and ExcluIR), TRIX improves instruction-aware ranking across sparse and dense frozen retrievers. It raises instruction sensitivity by up to 16.4 points and outperforms the strongest reported training-free method on exclusion-aware retrieval by up to 3.3 points.

Our contributions are:

- We introduce a training-free reranking framework that uses full, affirmative, and exclusion views to handle positive criteria, background context, and exclusions without retriever updates or document re-encoding.
- We use residual correction to obtain a query-local proxy for exclusion evidence beyond topical match, then combine relevance promotion and exclusion suppression according to this signal.
- Across three benchmarks and multiple frozen retrievers, we demonstrate consistent improvements and isolate the effects of view construction, residual scoring, and final score correction, while maintaining general retrieval effectiveness on eight BEIR datasets.

2 Related Work

Prior work on instruction control in retrieval falls into three groups: instruction-trained retrieval, inference-time query adaptation, and logic-based constraint modeling.

2.1 Instruction-Trained Retrieval

Instruction-trained retrievers learn to score documents according to natural-language instructions. TART uses multi-task instruction tuning with natural-language task descriptions (Asai et al. 2023). INSTRUCTOR learns a shared instruction-conditioned embedding space (Su et al. 2023). HypeR composes task-dependent prompts for cross-domain retrieval (Cai et al. 2023). More recent systems train directly with relevance instructions, including FollowIR-7B (Weller et al. 2025a), Promptriever (Weller et al. 2025b), and InF-IR (Zhuang et al. 2025). Dual-View Training adds polarity-reversed supervision (Zeng et al. 2026), while InstEmb combines input-intrinsic and output-aware signals when learning instruction-following embeddings (Gao et al. 2026).

These training-based methods require instruction-specific data and updates to the retriever parameters. Targeted negation augmentation likewise requires additional training data (Singh, Kumar, and Sridhar 2023), and supporting new instruction types can require further data and retraining. TRIX keeps the retriever and its representation function fixed and introduces instruction control during reranking.

2.2 Inference-Time Query Adaptation

Inference-time methods modify the query text or representation without retraining the retriever (Ma et al. 2023). HyDE generates a hypothetical document to guide retrieval (Gao et al. 2023). Query2Doc expands the query with a generated pseudo-document (Wang, Yang, and Wei 2023). RAG-Fusion combines rankings from multiple generated queries (Raudaschl 2024). Stage-Aware Decomposition retains the complete query for initial retrieval and additively aggregates decomposed condition scores during reranking (Yin et al. 2026). These methods add or reformulate topical content but do not handle exclusion constraints. Excluded entities can therefore remain strong lexical or semantic matching signals.

Other methods directly modify or construct query representations. DEO derives positive and negative subqueries and optimizes the query embedding toward the former and away from the latter (Lee et al. 2026). Because the resulting vector is not generated by the retriever's query encoder, it may deviate from the learned embedding structure and make its similarity to document embeddings less reliable. In vision–language retrieval, SpaceVLM-DRC uses angular composition to construct three negation-aware query directions and combines their image similarities with a direct image-space repulsion term (Pokhrel and Sheshappanavar 2026). This design assumes a unit-normalized shared image–text embedding space and treats direct similarity to the negated concept as evidence that an image contains it. Text retrieval therefore requires a scoring rule that supports sparse retrievers and distinguishes exclusion evidence from similarity caused by shared topic information.

TRIX formulates instruction control as score correction over the retrieved documents. It retains the complete query for candidate retrieval, keeps the query and document representations unchanged, estimates exclusion evidence from the relation between affirmative and exclusion scores, and combines relevance promotion with exclusion suppression. This score-based formulation applies to sparse and dense text retrievers.

2.3 Logic-Based Constraint Modeling

Logic-based methods represent constraints explicitly before reranking. NS-IR translates queries and documents into first-order logic and combines semantic alignment with connective constraints for negative-constraint retrieval (Xu et al. 2025). OrLog decomposes entity-seeking queries into atomic predicates and a logical form, then uses predicate plausibility scores in probabilistic logic inference (Hoveyda et al. 2026).

These methods enforce constraint composition through formal representations and candidate-level inference. The required logical translation, model calls, and formal inference increase online latency and make retrieval depend on the quality of the intermediate logical representation. TRIX operates directly on the scores of a frozen retriever, without translating documents into logical forms or invoking an external inference engine.

3 Methodology

Figure 1 presents the three stages of TRIX. Section 3.1 constructs full, affirmative, and exclusion evidence views. Section 3.2 computes a residual score as a query-local proxy for exclusion evidence beyond topical match. Section 3.3 uses this score to promote affirmative relevance and suppress strong exclusion matches.

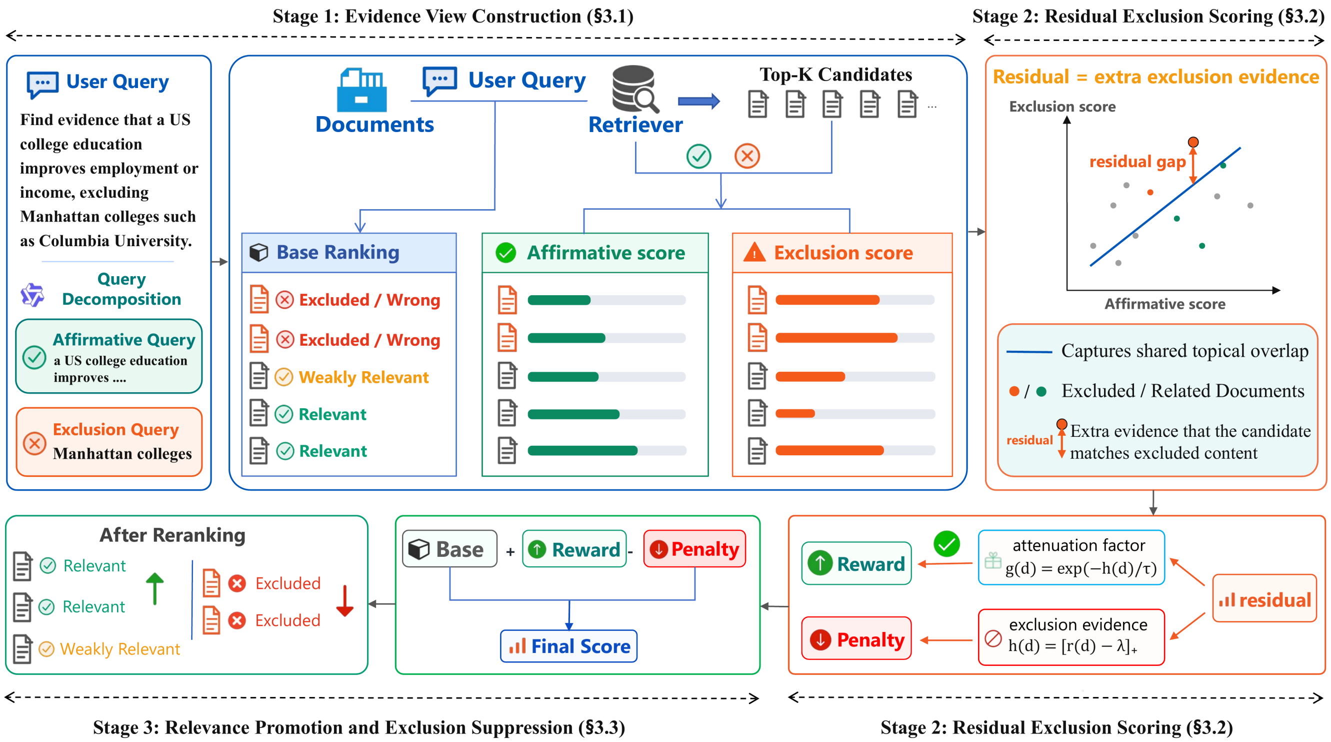

Figure 1: Overview of TRIX. Stage 1, Evidence View Construction (Section 3.1), constructs full, affirmative, and exclusion views and retrieves candidates with the full view. Stage 2, Residual Exclusion Scoring (Section 3.2), forms a query-local residual after removing shared score variation over the retrieved documents. Stage 3, Relevance Promotion and Exclusion Suppression (Section 3.3), converts this evidence into the final ranking correction.

3.1 Evidence View Construction

Stage 1 in Figure 1 constructs full, affirmative, and exclusion views while retaining the complete request for initial retrieval. The base ranking shows why the auxiliary views are needed: two documents about excluded Manhattan colleges are ranked first because they match both the desired college-employment topic and the excluded entities named in the request. The retriever treats both forms of lexical or embedding similarity as positive signals without reliably distinguishing their roles in the instruction.

TRIX receives a natural-language request u containing the retrieval topic and its instructions. The frozen retriever uses the complete request as the full query $Q_{\text{full}} = u$ and returns the top-K candidate set $\mathcal{C}_u$. An LLM decomposes u once into two auxiliary queries: Q_{pos} states the content that supports relevance, and Q_{neg} states the content to exclude. If the request contains no exclusion, the decomposer returns $Q_{\text{neg}} = $ [NONE]. The candidate set is always retrieved with Q_{full}; the auxiliary views are used only for reranking.

The three views serve different roles. The full query retains the complete topic, context, and instruction during candidate generation. The affirmative view keeps the target topic and positive relevance requirements while removing the exclusion clause and background information that does not determine relevance. The exclusion view states the rejected content as a positive description, allowing the retriever to score whether a document matches that content. In addition to supporting relevance, the affirmative score provides a reference for the topical component shared with the exclusion score. All views use the same frozen scorer.

Let $s(Q, d)$ denote the score assigned to document d under query Q. For each $d \in \mathcal{C}_u$, the three channels are

$$
\begin{aligned}
S_{\text{full}}(d) &= s(Q_{\text{full}}, d), \\
S_{\text{pos}}(d) &= s(Q_{\text{pos}}, d), \\
S_{\text{neg}}(d) &= s(Q_{\text{neg}}, d).
\end{aligned}
\tag{1}
$$

For a dense retriever, $s(Q_x, d) = \cos(f_q(Q_x), f_d(d))$ is the embedding similarity; for BM25, it is the lexical retrieval score. Dense document embeddings are computed once and reused across views, while BM25 uses one fixed index.

Table 1 illustrates the decomposition. The affirmative view retains the desired content, and the exclusion view converts the rejected content into a directly scorable description.

3.2 Residual Exclusion Scoring

Stage 2 in Figure 1 uses the residual as a query-local proxy for exclusion evidence beyond topical match. This avoids strongly penalizing candidates whose exclusion scores are high only because they closely match the topic. In the score plot, the fitted line represents the exclusion score expected from a candidate's affirmative score. The fifth, relevant candidate has high scores in both views but lies close to the fit-

View	Example text
Q_{full}	Find evidence that a US college education improves employment prospects or income. Exclude documents about colleges in Manhattan, such as Columbia University.
Q_{pos}	Evidence that a US college education improves employment prospects or income.
Q_{neg}	Colleges in Manhattan, such as Columbia University.

Table 1: Evidence views for the running example in Figure 1. The affirmative view states the desired evidence, while the exclusion view states the content to be excluded.

ted line, indicating that its exclusion score is expected from the shared college background. The highlighted orange candidate lies farther above the line and therefore produces a larger positive residual, indicating additional evidence that it matches the excluded Manhattan colleges.

TRIX addresses this ambiguity by estimating how much exclusion similarity is expected from the affirmative score and using the deviation from this relation as an exclusion proxy. Because all candidates in $\mathcal{C}_u$ are retrieved for the same request, the two scores can rise together when both respond to the shared topic. An affine fit provides a simple per-request estimate of this score dependence. The intercept captures a common offset, and the slope captures how the exclusion score changes with the affirmative score. If the two score channels have no systematic relation, the estimated slope is expected to be near zero. The fitted value then becomes approximately constant, so the standardized residual reduces to the normalized exclusion score up to centering and scaling.

TRIX first robustly standardizes all three score channels over the retrieved set $\mathcal{C}_u$:

$$z_x(d) = \frac{S_x(d) - \operatorname{median}_{d' \in \mathcal{C}_u} S_x(d')}{\operatorname{MAD}_{d' \in \mathcal{C}_u}(S_x(d')) + \epsilon}, \quad x \in \{\text{full}, \text{pos}, \text{neg}\}, \tag{2}$$

where $\operatorname{MAD}(S) = \operatorname{median}|S - \operatorname{median}(S)|$ and $\epsilon > 0$ prevents division by zero. This normalization aligns query-specific offsets and scales while limiting the influence of extreme scores (Rousseeuw and Croux 1993). TRIX then fits the affine relation with Huber regression:

$$(\hat{a}_u, \hat{b}_u) = \arg\min_{a,b} \sum_{d \in \mathcal{C}_u} \rho_\delta(z_{\text{neg}}(d) - a - b z_{\text{pos}}(d)), \tag{3}$$

where ρ_δ is the Huber loss with transition point δ. The fit captures the dominant score association, while the Huber loss limits the influence of large deviations (Huber 1964). It provides a query-specific reference for exclusion similarity. If the unnormalized positive-score channel has $\operatorname{MAD}(S_{\text{pos}}) < \epsilon$, we set $\hat{b}_u = 0$ and use a constant fitted value. For $e(d) = z_{\text{neg}}(d) - \hat{a}_u - \hat{b}_u z_{\text{pos}}(d)$, we robustly standardize the residual:

$$r(d) = \frac{e(d) - \operatorname{median}_{d' \in \mathcal{C}_u} e(d')}{\operatorname{MAD}_{d' \in \mathcal{C}_u}(e(d')) + \epsilon}. \tag{4}$$

The second standardization makes residual magnitudes comparable across requests: $r(d) = 0$ is the median fitted deviation, and one unit is measured relative to the within-request residual dispersion. This common scale allows the same decision parameters to be used across queries and retrievers. Positive $r(d)$ indicates more exclusion similarity than predicted by the document's affirmative score; negative $r(d)$ indicates less. The residual is a relative score-space proxy, not a calibrated probability of constraint violation. As detailed in the next stage, a residual value above the shared boundary reduces the affirmative reward and activates an exclusion penalty. Numerical details are given in Appendix E.

3.3 Relevance Promotion and Exclusion Suppression

Stage 3 in Figure 1 uses the residual proxy to control the final score correction. Candidates with strong affirmative support and a small residual retain a positive correction, while candidates with a large positive residual lose this support and receive a penalty. This raises relevant candidates and pushes the two initially top-ranked but excluded documents downward. We implement this correction by first retaining the positive part of the normalized affirmative score:

$$p(d) = [z_{\text{pos}}(d)]_+. \tag{5}$$

Because $z_{\text{pos}}(d)$ is median-centered, $p(d)$ rewards only documents whose affirmative score is above the candidate-set median. We activate the exclusion correction beyond a shared boundary $\lambda \geq 0$:

$$h(d) = [r(d) - \lambda]_+, \qquad g(d) = \exp\left(-\frac{h(d)}{\tau}\right), \quad \tau > 0, \tag{6}$$

where $h(d)$ is the positive residual excess above the boundary and $g(d) \in (0, 1]$ is the reward attenuation factor. Below the boundary, $h(d) = 0$ and $g(d) = 1$; above it, $h(d)$ grows linearly while $g(d)$ decreases smoothly. The final score is

$$S_{\text{final}}(d) = z_{\text{full}}(d) + p(d)g(d) - h(d). \tag{7}$$

The term $p(d)g(d)$ retains the affirmative reward below the residual boundary and attenuates it once the boundary is crossed. The term $-h(d)$ directly suppresses candidates with a large positive residual, so the correction can lower a document below its full-query score. All three terms use unit coefficients on their robust standardized scales, avoiding dataset- or retriever-specific scoring weights. For fixed $z_{\text{full}}(d)$ and $p(d)$, Eq. 7 is therefore non-increasing in $r(d)$. When no exclusion view is present, TRIX sets $h(d) = 0$ and $g(d) = 1$, reducing the score to $z_{\text{full}}(d) + p(d)$.

4 Experiments

4.1 Experimental Setup

Benchmarks and Metrics. FollowIR (Weller et al. 2025a) measures instruction-following retrieval on Core17, Robust04, and News21. We report aggregate retrieval effectiveness (Score) and p-MRR for instruction sensitivity. For documents whose relevance labels differ between paired instructions, p-MRR measures whether their reciprocal ranks move in the required direction; positive, zero, and negative values indicate correct, no, and incorrect rank changes, respectively. NegConstraint (Xu et al. 2025) and ExcluIR (Zhang et al. 2025) evaluate exclusion-aware retrieval in two settings. NegConstraint ranks a corpus and reports MAP and nDCG@10. ExcluIR compares desired–excluded document pairs using MRR and Right Rank (RR), the percentage of pairs ordered correctly. We further use eight datasets from BEIR (Thakur et al. 2021) to measure whether TRIX preserves standard retrieval effectiveness, reporting nDCG@10 and MAP@100.

Models and Settings. The frozen Qwen3-4B model (Yang et al. 2025) produces Q_{pos} and Q_{neg} in one structured generation on a single NVIDIA GeForce RTX 3090 GPU with temperature 0.1; Q_{full} remains the original request and retrieves the candidates. FollowIR experiments use BM25 (Robertson and Zaragoza 2009) together with the frozen E5-Mistral (Wang et al. 2024), BGE-large (Xiao et al. 2024), and RepLLaMA (Ma et al. 2024) bi-encoders. NegConstraint and ExcluIR use BGE-small-en-v1.5, BGE-large-en-v1.5, BGE-M3 (Xiao et al. 2024), and RepLLaMA. FollowIR, NegConstraint, and BEIR use $K_{\text{retrieve}} = K_{\text{fit}} = K_{\text{rerank}} = 1000$: Q_{full} retrieves 1,000 candidates, which form $\mathcal{C}_u$ for normalization, Huber fitting, and reranking. ExcluIR uses its full 90,406-document corpus because some annotated excluded documents initially rank near 9,000; this protocol keeps both documents available for every evaluated pair. Within every matched comparison, variants share candidates, generated views, retrieval templates, scorer configuration, and document representations. The same λ=0.5 and τ=0.1 are fixed without test relevance labels and used across datasets and retrievers. Appendix Table 10 reproduces the main FollowIR depth of 1,000 and reports its matched depth sensitivity analysis; this sweep does not alter the full-corpus ExcluIR protocol. Prompt, decoding, normalization, and numerical details are provided in Appendix E.

Baselines and Controlled Comparisons. On FollowIR, FollowIR-7B (Weller et al. 2025a), Promptriever (Weller et al. 2025b), and InstEmb (Gao et al. 2026) are included as instruction-trained baselines. We also reproduce HyDE (Gao et al. 2023), Query2Doc (Wang, Yang, and Wei 2023), and ConvSearch-R1 (Jiang et al. 2025) with RepLLaMA. Their aggregate comparison is reported in Section 4.2. On the exclusion benchmarks, NS-IR (Xu et al. 2025), DEO (Lee et al. 2026), and SEAL (Zhang et al. 2025) are included as prior methods. Their results are separated from the matched Base/+TRIX comparisons because their training and retrieval pipelines differ.

4.2 Main Results on Instruction-Following Retrieval

Table 2 reports aggregate FollowIR results. The lower block contains matched Base/+TRIX pairs spanning BM25 and three dense retriever families; the upper block reports instruction-trained retrievers. Per-dataset results are reported in Appendix A.

TRIX improves both retrieval effectiveness and instruction sensitivity for every frozen backbone, spanning BM25 and three dense retrievers. Before reranking, the three dense backbones have negative average p-MRR, indicating that their rankings often fail to follow the relevance instruction. TRIX makes p-MRR positive while also raising Score; with RepLLaMA, Score increases from 23.8 to 28.7 and p-MRR from -3.3 to $+12.3$. Each Base/+TRIX pair ranks the same candidates with the same document representations; TRIX only corrects their scores.

Method	**Score**	**p-MRR**
Instruction-trained retrievers		
FollowIR-7B	24.8	+12.2
Promptriever	26.1	+11.2
InstEmb (Llama-3-8B)	**28.5**	**+15.6**
Matched frozen-backbone comparisons		
BM25	15.2	+1.1
+TRIX	17.3	**+17.5**
E5-Mistral	26.1	−2.7
+TRIX	28.0	+12.1
BGE-large	19.2	−2.2
+TRIX	23.3	+6.5
RepLLaMA	23.8	−3.3
+TRIX	**28.7**	+12.3

Table 2: Aggregate FollowIR results. Score averages MAP on Core17 and Robust04 with nDCG on News21. Each +TRIX row is matched to the backbone above it; values are percentages ($\times 100$). Best results within each comparable block are shown in bold.

Method	**Score**	**p-MRR**
RepLLaMA	23.8	−3.3
HyDE	23.4	+1.4
Query2Doc	25.6	−1.1
ConvSearch-R1	23.4	+0.2
TRIX	**28.7**	**+12.3**

Table 3: Aggregate FollowIR comparison with query rewriting and expansion using RepLLaMA. Score averages MAP on Core17 and Robust04 with nDCG on News21; values are percentages ($\times 100$). Best results are shown in bold.

Comparison with Query Rewriting and Expansion. Table 3 compares TRIX with representative inference-time rewriting and expansion methods using RepLLaMA. Each method uses its original query-generation and retrieval procedure. Results for Core17, Robust04, and News21 are provided in Appendix Table 8.

TRIX outperforms all compared rewriting and expansion methods on both aggregate metrics, reaching 28.7 Score and $+12.3$ p-MRR. Existing rewriting methods are mainly designed to make the target topic easier to match. For example, HyDE generates a hypothetical relevant document, while Query2Doc expands the query with a generated pseudo-document. Both add positive topical context, but neither represents excluded content as evidence that should lower a document's rank. Excluded entities can therefore remain

Backbone / Method	NegConstraint		ExcluIR	
	MAP	nDCG@10	MRR	RR
Prior methods				
NS-IR (GPT-4o)	53.3	56.5	—	—
DEO (BGE-large)	**73.3**	**78.8**	74.1	**79.3**
SEAL (trained)	—	—	**78.4**	78.0
Matched frozen-backbone comparisons				
BGE-small	68.4	75.2	75.0	60.1
+TRIX	76.9	82.3	77.1	81.3
BGE-large	63.8	71.9	73.4	56.0
+TRIX	76.6	81.8	78.2	82.2
BGE-M3	64.3	74.1	78.9	67.0
+TRIX	74.7	81.1	79.7	83.7
RepLLaMA	72.8	79.4	75.9	62.1
+TRIX	**80.0**	**84.8**	**87.2**	**89.9**

Table 4: Results on NegConstraint and ExcluIR. NegConstraint uses top-1,000 candidates, whereas ExcluIR scores its full 90,406-document corpus. Each +TRIX row uses the same candidate pool and document representations as its base; values are percentages ($\times 100$). Best results within each comparable block are shown in bold.

strong lexical or semantic matching signals after rewriting. TRIX separates affirmative and exclusion evidence before correcting the ranking scores.

4.3 Evaluation on Exclusion-Aware Benchmarks

We evaluate TRIX on two benchmarks designed for retrieval with negative constraints. NegConstraint (Xu et al. 2025) evaluates compositional constraints over a ranked corpus, while ExcluIR (Zhang et al. 2025) compares desired and excluded documents in pairs.

NegConstraint reranks the top 1,000 full-query candidates. ExcluIR scores the complete 90,406-document corpus with all three views and uses this pool for normalization, Huber fitting, and reranking. The full pool covers annotated excluded documents that initially rank near 9,000. Regression uses only retrieval scores and no pair labels. MRR and RR are computed from the resulting full rankings under the protocol detailed in Appendix C.

TRIX improves both metrics for every matched backbone on both benchmarks. The gains hold across the three BGE models and RepLLaMA, showing that the same reranking rule is effective across different retrievers and in both corpus ranking and pairwise comparison. RepLLaMA provides the strongest overall results: TRIX raises its MAP from 72.8 to 80.0 and nDCG@10 from 79.4 to 84.8 on NegConstraint, while improving MRR from 75.9 to 87.2 and RR from 62.1 to 89.9 on ExcluIR.

Compared with prior methods, RepLLaMA with TRIX obtains higher results than DEO on both benchmarks and SEAL on ExcluIR. These methods use different training and retrieval pipelines, so the results are not controlled comparisons. Appendix C compares raw exclusion scoring with residual scoring on both benchmarks.

Method	Avg. nDCG@10	Avg. MAP@100
BGE-large	60.03	44.23
+TRIX	60.22	44.19

Table 5: General retrieval performance on eight BEIR datasets with BGE-large. TRIX maintains average retrieval effectiveness, with changes of $+0.19$ nDCG@10 and -0.04 MAP@100; values are percentages ($\times 100$).

Variant	Decision score	Score	p-MRR
Evidence construction			
Full-query baseline	z_{full}	23.8	-3.3
Positive-view scoring	$z_{\text{full}} + p$	25.5	-0.7
Exclusion-view subtraction	$z_{\text{full}} - z_{\text{neg}}$	25.8	+7.6
Exclusion-signal construction			
Raw exclusion subtraction	$z_{\text{full}} + p - z_{\text{neg}}$	24.0	+14.9
Residual exclusion subtraction	$z_{\text{full}} + p - r$	26.6	+12.7
TRIX score composition			
TRIX w/o reward attenuation	$z_{\text{full}} + p - h$	26.3	+10.8
TRIX w/o exclusion penalty	$z_{\text{full}} + pg$	26.9	+8.9
Full TRIX	$z_{\text{full}} + pg - h$	**28.7**	**+12.3**

Table 6: Controlled mechanism ablation on FollowIR. The three blocks isolate evidence construction, exclusion-signal construction, and the two operations in TRIX score composition. Score averages MAP on Core17 and Robust04 with nDCG on News21; all values are percentages ($\times 100$). Results for the full system are shown in bold.

4.4 General Retrieval Performance

We use eight BEIR datasets to measure general retrieval performance. Table 5 reports aggregate results with BGE-large; the per-dataset breakdown is provided in Appendix F.

TRIX maintains general retrieval performance across the eight BEIR datasets. Average nDCG@10 improves from 60.03 to 60.22, while MAP@100 remains nearly unchanged (44.23 versus 44.19), corresponding to changes of $+0.19$ and -0.04 points, respectively.

4.5 Ablation Study

Table 6 organizes the ablation around three design steps: evidence view construction, residual exclusion scoring, and final score correction. Here p, r, h, and g denote the positive score, residual exclusion score, thresholded exclusion penalty, and reward attenuation factor defined in Section 3. All variants use the same candidates, generated views, frozen encoder, templates, embeddings, and hyperparameters.

Evidence Views. The focused positive view raises Score from 23.8 to 25.5 and p-MRR from -3.3 to -0.7, confirming that it contributes evidence beyond the full instruction. The exclusion view also contributes independently: subtracting its score raises Score to 25.8 and p-MRR to $+7.6$. The two views therefore provide complementary signals for relevance promotion and exclusion suppression.

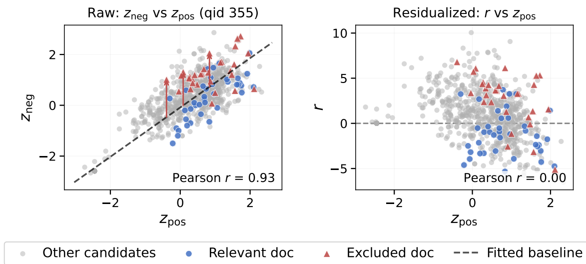

Figure 2: Residualization for an illustrative query. Left: the Huber relation between affirmative and exclusion scores. Right: standardized deviations after removing the fitted component. Positive residuals identify candidates with more exclusion similarity than predicted by their affirmative score.

Residual Exclusion Evidence. Under the same subtraction rule, replacing raw exclusion similarity with the residual recovers 2.6 Score points while retaining strongly positive p-MRR (12.7 versus 14.9). Raw subtraction causes large downward rank shifts for many documents with high exclusion scores. This produces high p-MRR because more documents are demoted and move farther down the ranking. A high exclusion score, however, does not necessarily indicate excluded content: some documents that satisfy the instruction also receive high exclusion scores because they closely match its topic. Direct subtraction over-penalizes these documents and limits Score to 24.0. The residual provides a query-local proxy for exclusion evidence beyond topical match, reducing the over-penalization of documents that satisfy the instruction. It raises Score to 26.6 while retaining a positive p-MRR of 12.7.

Score Correction. Full TRIX improves over the variant without reward attenuation by 2.4 Score and 1.5 p-MRR points, showing that strong residual evidence should also reduce the positive contribution. Adding the explicit penalty to the attenuation-only variant yields gains of 1.8 Score and 3.4 p-MRR points. Together, the two comparisons support the asymmetric combination of attenuating affirmative support and directly suppressing strong exclusion matches.

4.6 Mechanism Analysis

Residual Signal. Figure 2 illustrates how residualization changes the meaning of the exclusion score. For this query, z_{neg} increases with z_{pos} because the affirmative and exclusion views share topical information. The fitted relation represents the exclusion score expected from this shared relevance. A document close to the fitted relation may therefore have a high exclusion score without providing strong evidence that it matches the excluded content.

Candidates near the fitted relation receive small residuals, so high exclusion scores explained by topical match do not trigger a strong penalty. Candidates above the relation receive positive residuals, indicating additional exclusion evidence beyond topical match. This distinction explains the controlled result in Table 6: replacing raw exclusion subtraction with the residual raises Score from 24.0 to 26.6 while retaining

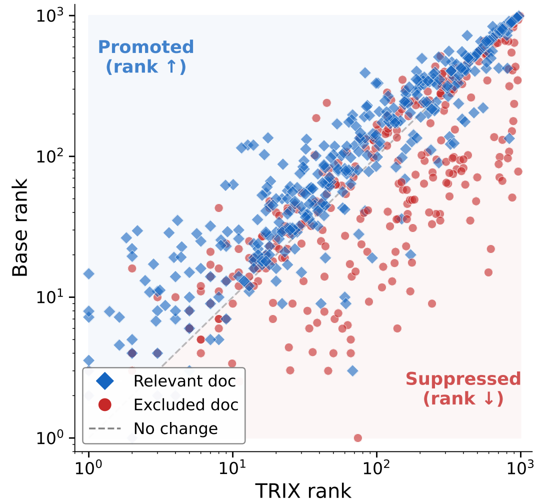

Figure 3: Rank changes produced by TRIX on FollowIR exclusion queries. The vertical axis gives the initial rank from the frozen retriever, and the horizontal axis gives the rank after TRIX adds the affirmative reward and subtracts the exclusion penalty. Points above the diagonal are promoted, points below it are suppressed, and points on the diagonal are unchanged. Blue diamonds denote relevant documents, while orange circles denote documents that violate the exclusion constraint.

a strongly positive p-MRR of $+12.7$. The residual reduces penalties on relevant documents whose raw exclusion scores are high mainly because of topic overlap.

Ranking Behavior. Figure 3 visualizes the document-level rank changes produced by TRIX.

The distribution shows how the correction acts on the two document groups. Relevant documents, shown as blue diamonds, are concentrated above the diagonal, indicating that TRIX promotes documents that should rank higher. Documents that violate the exclusion constraint tend to fall below the diagonal, indicating that the correction suppresses documents that should rank lower. This separation is clearest among the highest-ranked documents, where rank changes have the largest effect on retrieval quality. Together with the score-composition ablation in Table 6, this pattern illustrates how the affirmative reward and exclusion penalty jointly produce the intended ranking behavior.

5 Conclusion and Limitations

This paper presented TRIX, a training-free method that corrects the scores of a frozen retriever to better follow complex relevance instructions without retraining or document re-encoding. The results support three conclusions. First, separating the request into focused evidence views improves relevance scoring for positive criteria. Second, residual scoring provides a query-local proxy for exclusion evidence beyond topical match and reduces the over-penalization of documents that satisfy the instruction. Third, using this signal to attenuate affirmative rewards and apply exclusion penalties improves instruction-sensitive ranking across the matched sparse and dense frozen retrievers. On eight BEIR datasets, TRIX maintains general retrieval effectiveness, with average changes of $+0.19$ nDCG@10 and -0.04 MAP@100 relative to BGE-large.

TRIX relies on two conditions: the LLM decomposition must expose useful affirmative and exclusion views, and the initial retriever must provide adequate candidate coverage. As a reranker, TRIX cannot recover relevant documents absent from the initial top-K set. The reported results evaluate score correction under full-query candidate generation; alternative candidate-generation strategies form a separate retrieval-stage design choice. Its local affine correction may also be less effective when the two views remain tightly coupled in the retriever's score space. Future work can combine score correction with stronger decomposition and explicit constraint reasoning.

References

Asai, A.; Schick, T.; Lewis, P.; Chen, X.; Izacard, G.; Riedel, S.; Hajishirzi, H.; and Yih, W. 2023. Task-aware Retrieval with Instructions. In Rogers, A.; Boyd-Graber, J. L.; and Okazaki, N., eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, volume ACL 2023 of *Findings of ACL*, 3650–3675. Association for Computational Linguistics.

Cai, Z.; Tao, C.; Shen, T.; Xu, C.; Geng, X.; Lin, X. A.; He, L.; and Jiang, D. 2023. HypeR: Multitask Hyper-Prompted Training Enables Large-Scale Retrieval Generalization. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.

Gao, L.; Ma, X.; Lin, J.; and Callan, J. 2023. Precise Zero-Shot Dense Retrieval without Relevance Labels. In *ACL*, 1762–1777.

Gao, T.; Fang, J.; Zhang, X.; Liu, Z.; Liu, C.; Liu, P.; and Jiang, Q. 2026. InstEmb: Instruction-Following Embeddings through Glimpses of the Future. In *Proceedings of the 43rd International Conference on Machine Learning*.

Hoveyda, M.; Piepenbrock, J.; de Vries, A. P.; de Rijke, M.; and Hasibi, F. 2026. OrLog: Resolving Complex Queries with LLMs and Probabilistic Reasoning. In Campos, R.; Jatowt, A.; Lan, Y.; Aliannejadi, M.; Bauer, C.; MacAvaney, S.; Anand, A.; Ren, Z.; Verberne, S.; Bai, N.; and Mansoury, M., eds., *Advances in Information Retrieval - 48th European Conference on Information Retrieval, ECIR 2026, Delft, The Netherlands, March 29 - April 2, 2026, Proceedings, Part I*, volume 16483 of *Lecture Notes in Computer Science*, 98–114. Springer.

Huber, P. J. 1964. Robust Estimation of a Location Parameter. *The Annals of Mathematical Statistics*, 35(1): 73–101.

Jiang, P.; Lin, J.; Cao, L.; Tian, R.; Kang, S.; Wang, Z.; Sun, J.; and Han, J. 2025. DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning. *CoRR*, abs/2503.00223.

Lee, T.; Park, J.; Hwang, S.; and Jang, J. 2026. DEO: Training-Free Direct Embedding Optimization for Negation-Aware Retrieval. *CoRR*, abs/2603.09185.

Ma, X.; Gong, Y.; He, P.; Zhao, H.; and Duan, N. 2023. Query Rewriting in Retrieval-Augmented Large Language Models. In *EMNLP*.

Ma, X.; Wang, L.; Yang, N.; Wei, F.; and Lin, J. 2024. Fine-Tuning LLaMA for Multi-Stage Text Retrieval. In *SIGIR*, 2421–2425.

Pokhrel, A.; and Sheshappanavar, S. V. 2026. Negation Matters: Training-Free Negation-Aware Image Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 11353–11362.

Ralev, R.; Baral, A.; Zhechev, I.; Agarwal, J.; and Rajamohan, S. 2026. Training for Compositional Sensitivity Reduces Dense Retrieval Generalization. *CoRR*, abs/2604.16351.

Raudaschl, S. 2024. RAG-Fusion: A New Take on Retrieval-Augmented Generation.

Robertson, S. E.; and Zaragoza, H. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4): 333–389.

Rousseeuw, P. J.; and Croux, C. 1993. Alternatives to the Median Absolute Deviation. *Journal of the American Statistical Association*, 88(424): 1273–1283.

Singh, R.; Kumar, R.; and Sridhar, V. 2023. NLMs: Augmenting Negation in Language Models. In Bouamor, H.; Pino, J.; and Bali, K., eds., *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, volume EMNLP 2023 of *Findings of ACL*, 13104–13116. Association for Computational Linguistics.

Su, H.; Shi, W.; Kasai, J.; Wang, Y.; Hu, Y.; Ostendorf, M.; Yih, W.; Smith, N. A.; Zettlemoyer, L.; and Yu, T. 2023. One Embedder, Any Task: Instruction-Finetuned Text Embeddings. In Rogers, A.; Boyd-Graber, J. L.; and Okazaki, N., eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, volume ACL 2023 of *Findings of ACL*, 1102–1121. Association for Computational Linguistics.

Thakur, N.; Reimers, N.; Rücklé, A.; Gurevych, I.; and Daxenberger, J. 2021. BEIR: A Heterogeneous Benchmark for Zero-Shot Evaluation of Information Retrieval Models. In *NeurIPS Datasets and Benchmarks*.

Wang, L.; Yang, N.; Huang, X.; Yang, L.; Majumder, R.; and Wei, F. 2024. Improving Text Embeddings with Large Language Models. In *ACL*, 11897–11916.

Wang, L.; Yang, N.; and Wei, F. 2023. Query2doc: Query Expansion with Large Language Models. In *EMNLP*, 9414–9423.

Weller, O.; Chang, B.; MacAvaney, S.; Lo, K.; Cohan, A.; Durme, B. V.; Lawrie, D.; and Soldaini, L. 2025a. FollowIR: Evaluating and Teaching Information Retrieval Models to Follow Instructions. In *NAACL*, 11926–11942.

Weller, O.; Durme, B. V.; Lawrie, D.; Paranjape, A.; Zhang, Y.; and Hessel, J. 2025b. Promptriever: Instruction-Trained Retrievers Can Be Prompted Like Language Models. In *ICLR*.

Weller, O.; Lawrie, D. J.; and Durme, B. V. 2024. NevIR: Negation in Neural Information Retrieval. In Graham, Y.; and Purver, M., eds., *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics, EACL 2024 - Volume 1: Long Papers, St. Julian's, Malta, March 17-22, 2024*, 2274–2287. Association for Computational Linguistics.

Xiao, S.; Liu, Z.; Zhang, P.; Muennighoff, N.; Lian, D.; and Nie, J. 2024. C-Pack: Packed Resources For General Chinese Embeddings. In *SIGIR*, 641–649.

Xu, G.; Zhang, Z.; Mei, W.; Liang, J.; Lu, W.; Zhang, X.; Yang, Z.; Ma, X.; Xiao, Y.; and Yang, D. 2025. Logical Consistency is Vital: Neural-Symbolic Information Retrieval for Negative-Constraint Queries. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics: ACL 2025*, 1828–1847. Vienna, Austria: Association for Computational Linguistics.

Yang, A.; Li, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Gao, C.; Huang, C.; Lv, C.; et al. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*.

Yin, B.; Lu, X.; Qi, Z.; and Shen, X. 2026. When Should Queries Be Decomposed? A Stage-Aware Study of Query Decomposition for Multi-Condition Retrieval. *arXiv preprint arXiv:2606.08577*.

Zeng, Q.; et al. 2026. Dual-View Training for Instruction-Following Information Retrieval. *arXiv preprint arXiv:2604.18845*.

Zhang, W.; Zhang, M.; Wu, S.; Pei, J.; Ren, Z.; de Rijke, M.; Chen, Z.; and Ren, P. 2025. ExcluIR: Exclusionary Neural Information Retrieval. In *AAAI*, 13295–13303.

Zhuang, Y.; Trinh, A.; Qiang, R.; Sun, H.; Zhang, C.; Dai, H.; and Dai, B. 2025. Towards Better Instruction Following Retrieval Models. *arXiv preprint arXiv:2505.21439*.

A Full FollowIR Results

Table 7 provides the per-dataset results underlying the aggregate comparison in Table 2. Results for instruction-trained methods are taken from their source papers (Weller et al. 2025a,b; Gao et al. 2026).

(a) Retrieval effectiveness

Method	Core17	Robust04	News21	Score
Instruction-trained retrievers				
FollowIR-7B	20.0	24.8	29.6	24.8
Promptriever	21.6	28.3	28.5	26.1
InstEmb (Llama-3-8B)	24.0	29.2	32.3	28.5
Matched frozen-backbone comparisons				
BM25	10.8	13.8	21.0	15.2
+TRIX	13.2	14.0	24.8	17.3
E5-Mistral	20.4	27.9	30.1	26.1
+TRIX	24.4	26.7	32.8	28.0
BGE-large	16.8	19.4	21.4	19.2
+TRIX	22.1	21.9	26.0	23.3
RepLLaMA	23.2	25.7	22.5	23.8
+TRIX	25.9	28.2	32.1	28.7

(b) Instruction sensitivity (p-MRR)

Method	Core17	Robust04	News21	Average
Instruction-trained retrievers				
FollowIR-7B	+16.5	+13.7	+6.3	+12.2
Promptriever	+15.4	+11.7	+6.4	+11.2
InstEmb (Llama-3-8B)	+22.4	+19.1	+5.4	+15.6
Matched frozen-backbone comparisons				
BM25	+3.7	−0.5	0.0	+1.1
+TRIX	+18.4	+13.8	+20.3	+17.5
E5-Mistral	−0.3	−6.9	−0.9	−2.7
+TRIX	+5.7	+19.6	+11.1	+12.1
BGE-large	−2.0	−5.2	+0.5	−2.2
+TRIX	+4.3	+9.0	+6.3	+6.5
RepLLaMA	+1.1	−9.1	−1.8	−3.3
+TRIX	+16.2	+14.1	+6.8	+12.3

Table 7: Per-dataset FollowIR results. The two panels report retrieval effectiveness and p-MRR for the same systems; values are percentages ($\times 100$).

B Additional Ablations and Sensitivity

The main text isolates the positive view, residual exclusion signal, reward attenuation, and explicit penalty. The following results compare alternative residual estimators and examine sensitivity to the shared configuration.

B.1 Estimator and Normalization Choices

Table 9 compares the robust estimator, normalization, and residual construction used to obtain r.

Table 10 varies one factor at a time around the main RepLLaMA configuration. The $K = 1000$ row reproduces the main FollowIR result (28.7 Score and +12.3 p-MRR), resolving the candidate-depth setting used by Tables 2 and 3. All tested depths improve over the RepLLaMA base result of 23.8 Score and -3.3 p-MRR, although the magnitude is depth-sensitive: Score ranges from 26.3 to 28.7, while p-MRR varies non-monotonically from +12.3 to +22.5. Because the same candidate set is used for retrieval, robust normalization, Huber fitting, and reranking, this depth analysis measures the joint effect of all three stages. Separating candidate coverage from score-estimation effects requires holding retrieval depth fixed while varying the estimation pool. In the one-factor parameter sweeps, increasing either λ or τ reduces both reported metrics relative to the tested defaults.

(a) Retrieval effectiveness

Method	Core17	Robust04	News21	Score
RepLLaMA	23.2	25.7	22.5	23.8
HyDE	23.6	24.9	21.4	23.4
Query2Doc	25.9	26.1	24.9	25.6
ConvSearch-R1	22.3	25.9	22.0	23.4
TRIX	25.9	28.2	32.1	28.7

(b) Instruction sensitivity (p-MRR)

Method	Core17	Robust04	News21	Average
RepLLaMA	+1.1	−9.1	−1.8	−3.3
HyDE	+6.5	−2.9	+0.7	+1.4
Query2Doc	+8.0	−7.5	−3.8	−1.1
ConvSearch-R1	+3.0	−2.5	+0.2	+0.2
TRIX	+16.2	+14.1	+6.8	+12.3

Table 8: Per-dataset FollowIR comparison with query rewriting and expansion using RepLLaMA. The two panels report retrieval effectiveness and p-MRR; values are percentages ($\times 100$).

Variant	Modification	Score	p-MRR
Full TRIX	None	28.7	+12.3
TRIX w/ OLS	Replace Huber regression	26.9	+10.6
TRIX w/ mean/std scaling	Replace median/MAD scaling	26.8	+8.5
TRIX w/o residual centering	Omit residual recentering	27.1	+10.8

Table 9: Secondary TRIX design choices on FollowIR. Each row changes one estimation or normalization choice while retaining the controlled candidate set and three evidence views.

Instruction decomposition averages 2.81 seconds per query (standard deviation 1.20, range 1.51–6.80 seconds), corresponding to a throughput of 0.36 decomposition queries per second. After the three retrieval-score channels are available, score normalization, Huber fitting, and final reranking take 0.4 ms per query. This post-scoring measurement excludes query decomposition, query encoding, and candidate retrieval, and therefore is not an end-to-end latency measurement. Among the reported online components, latency is dominated by the one-time query decomposition. TRIX requires no retriever training, document re-encoding, or candidate-level LLM inference; its additional LLM cost is one query-level decomposition.

Analysis	Setting	Score	p-MRR
Threshold	$\lambda = 0.5$ (default)	28.7	+12.3
	$\lambda = 1.0$	28.3	+11.8
	$\lambda = 1.5$	28.1	+9.2
	$\lambda = 2.0$	27.8	+7.5
Temperature	$\tau = 0.1$ (default)	28.7	+12.3
	$\tau = 0.2$	28.2	+11.9
	$\tau = 0.5$	27.8	+10.4
	$\tau = 1.0$	27.1	+8.7
Candidate depth	$K = 10$	26.3	+18.1
	$K = 50$	27.9	+20.8
	$K = 100$	28.5	+22.5
	$K = 200$	28.4	+22.1
	$K = 1000$ (default)	28.7	+12.3
Query decomposition	Qwen3-4B	2.81 ± 1.20 s/q	
Post-scoring correction	Cached scores	0.4 ms/q	

Table 10: Sensitivity and efficiency with RepLLaMA on FollowIR. Each sensitivity block varies one factor while retaining the other defaults (K=1000, λ=0.5, and τ=0.1). Candidate depth K jointly specifies the number of Q_{full} candidates retrieved, used for normalization and Huber fitting, and reranked before metric computation. Score and p-MRR are averaged over the three FollowIR datasets. Decomposition latency is measured across all FollowIR queries using Qwen3-4B on a single NVIDIA GeForce RTX 3090 GPU with temperature 0.1; we report mean ± standard deviation. The 0.4 ms measurement covers score normalization, Huber fitting, and final reranking after retrieval scores are available.

C Additional Exclusion-Benchmark Results

Table 4 (main text, Section 4.3) reports the complete TRIX results across NegConstraint and ExcluIR. Table 11 examines whether its performance depends on using the per-query regression residual instead of raw exclusion similarity. The same matched comparison is applied to both benchmarks. Within each benchmark, all rows use the same frozen retriever, candidate pool, decomposition, query templates, document embeddings, and shared λ, τ. The pool contains the top 1,000 candidates for NegConstraint and the complete 90,406-document corpus for ExcluIR. The raw and residual correction rows differ only in the exclusion signal supplied to the same TRIX scoring function.

ExcluIR Evaluation Protocol. For an ExcluIR request u, let $\operatorname{rank}_u(d) \in \{1, \ldots, 90{,}406\}$ be the one-based rank of document d after reranking the full corpus. For an annotated desired document d^+, its top-10 reciprocal-rank contribution is $1/\operatorname{rank}_u(d^+)$ when $\operatorname{rank}_u(d^+) \leq 10$ and zero otherwise; MRR averages this quantity over the evaluation instances. For each annotated desired–excluded pair (d^+, d^-), Right Rank is one when $\operatorname{rank}_u(d^+) < \operatorname{rank}_u(d^-)$ and zero otherwise; RR is its mean over all pairs. Equal ranks contribute zero. Full-corpus scoring guarantees that neither pair member is missing from the evaluated ranking.

Result Provenance. The DEO NegConstraint values in Table 4 are taken from its source paper, while we reproduce its ExcluIR values under the benchmark protocol above. The SEAL ExcluIR values are taken from its source paper. These rows provide prior-method context; the Base/+TRIX blocks are the matched comparisons used to isolate TRIX's score correction.

Variant	NegConstraint		ExcluIR	
	MAP	nDCG@10	MRR	RR
Full-query baseline	63.8	71.9	73.4	56.0
TRIX w/ positive view only	76.0	81.8	75.0	58.0
TRIX w/ raw exclusion score	76.6	81.8	76.0	60.0
Full TRIX	76.6	81.8	78.2	82.2

Table 11: Matched exclusion-signal ablation across exclusion benchmarks with BGE-large-en-v1.5. Within each benchmark, all variants use the same candidate pool, generated views, document representations, and scoring hyperparameters.

The two benchmarks show different contributions from the exclusion signal. On NegConstraint, the positive view accounts for most of the gain over the full-query baseline: it raises MAP from 63.8 to 76.0 and nDCG@10 from 71.9 to 81.8. Raw exclusion scoring adds 0.6 MAP points and no nDCG@10 gain, while replacing it with the residual leaves both reported metrics unchanged at 76.6 and 81.8. On ExcluIR, the residual correction improves over raw exclusion scoring from 76.0 to 78.2 MRR and from 60.0 to 82.2 RR. Thus, in this matched comparison, residualization contributes most clearly to the desired–excluded pair ordering measured by ExcluIR, whereas the NegConstraint improvement is driven primarily by the positive view.

D Qualitative Analysis

The qualitative analysis contains one representative success and one weak-exclusion-signal failure from FollowIR (Core17) with RepLLaMA. Each case reports the benchmark, query ID, complete request, decomposed views, candidate scores, and the ranking change. The success case is the query with the largest single-document rank drop; the failure case is the query with the smallest rank change among those carrying an exclusion instruction.

E Decomposition and Reproducibility Details

The instruction decomposer is the frozen Qwen3-4B model (Yang et al. 2025). It receives the complete user request and returns structured affirmative and exclusion content in one generation. The full view Q_{full} is the original request itself, used unchanged for candidate generation; Q_{pos} retains the topic and affirmative conditions while removing exclusion content, and Q_{neg} contains the excluded content, or [NONE] when no exclusion is present. We use the fixed prompt with temperature 0.1, top-p = 0.9, at most 512 new tokens, and thinking disabled. Malformed or missing fields are mapped to the corresponding empty view. The system prompt is:

```
You are an expert IR query optimizer
for Dense Vector Models. Decouple
"Original Query" and "Instruction"
```

Outcome	Request and views	Candidate evidence	Ranking effect and interpretation
Success	Benchmark: FollowIR (Core17), QID 355 Query: "Identify documents discussing the development and application of space-borne ocean remote sensing" Exclusion: land-bound sciences, agriculture, forestry, marketing, temperature Views: Q_{full} (full query), Q_{pos} (ocean remote sensing in oceanography, seabed prospecting, marine sciences), Q_{neg} (land-bound sciences, agriculture, forestry, marketing, temperature)	Original top-1 doc ($s_{\text{full}} = 0.725$) scores high on both s_{pos} and s_{neg} channels (agriculture/forestry content). Residual r remains high after accounting for their shared score variation. After TRIX: top-1 score rises to 1.907 for ocean-only doc; land-bound doc drops to rank 580.	Land-bound/agriculture documents drop from top-10 to rank > 500. Ocean remote-sensing documents rise to top-5. **Interpretation:** TRIX uses the high residual to suppress the excluded content in this candidate set.
Failure	Benchmark: FollowIR (Core17), QID 310 Query: "Evidence that radio waves from radio towers or car (mobile) phones affect brain cancer occurrence" Exclusion: leukemia Views: Q_{full} (full query), Q_{pos} (radio waves and brain cancer incidence), Q_{neg} (leukemia)	Original top-1 doc ($s_{\text{full}} = 0.795$) discusses brain cancer broadly. s_{neg} channel for "leukemia" is low because leukemia-specific mentions are rare in the candidate set, yielding near-zero residual r for the top document. After TRIX: top-1 score rises to 1.689 but ranking is unchanged.	Top-5 ranking is unchanged; excluded-topic documents remain in place. **Interpretation:** When the exclusion term rarely co-occurs with the target in the candidate set, the residual is too small to trigger a penalty—the failure mode is low negative-channel signal rather than semantic entanglement.

Table 12: Representative TRIX cases from FollowIR Core17 with RepLLaMA. Each case reports the operational views, score-space evidence, and ranking change under the complete scoring rule.

```
into Q_plus and Q_minus.

Q_plus rules: (1) Fluent natural
language phrase, not a keyword list.
(2) Semantic topic only; remove format
noise ("articles", "documents",
"reports"). (3) Remove conversational
connectors ("such as", "including").
(4) Sharp, coherent.

Q_minus rules: (1) Extract only
critical high-level exclusion concepts.
(2) Concise natural phrasing.
(3) If no exclusion, output "[NONE]".

No-negation rule: DENSE RETRIEVERS
IGNORE LOGIC WORDS. NEVER use negation
in Q_minus ("no", "not", "non-",
"without"). Convert negations to
AFFIRMATIVE entities:
  BAD: "non-United Kingdom"
  -> GOOD: "United States, France, global"
  BAD: "violence outside Ireland"
  -> GOOD: "violence in England,
     international"
Output ONLY exact entities to exclude.

Output JSON:
{
  "Reasoning_Steps": "1. Core topic.
    2. Filter noise. 3. Exclusions
    (negations->affirm).",
  "Q_plus": "Fluent natural language query",
  "Q_minus": "Core exclusions or [NONE]"
}
```

The user prompt template is:

```
Decouple into Q_plus and Q_minus:
Query: "{query}"
Instruction: "{instruction}"
```

For every backbone, including RepLLaMA, all document representations are computed once with that retriever's native document template, and all three query views use its native query template. Matched Base/+TRIX variants reuse the same candidates, templates, and document representations. The $K_{\text{retrieve}} = K_{\text{fit}} = K_{\text{rerank}} = 1000$ convention in Section 4.1 applies to the reported FollowIR, NegConstraint, and BEIR results. ExcluIR uses the complete 90,406-document corpus as $\mathcal{C}_u$ for score normalization, Huber fitting, and reranking.

Numerical and Scoring Constants. We set the Huber transition point to $\delta = 1.35$ in standardized score units. The constant $\epsilon = 10^{-6}$ serves both as the median/MAD denominator floor and as the low-dispersion threshold: when $\mathrm{MAD}(S_{\text{pos}}) < \epsilon$, the fitted slope is set to zero. The regression is fit independently on each candidate set and uses no relevance labels. We use $\lambda = 0.5$ and $\tau = 0.1$ as global heuristic constants defined on the standardized residual scale and fix them across datasets and retrievers without per-dataset or per-retriever tuning. Because r is centered and scaled by its within-query MAD, $\lambda = 0.5$ activates the correction half a residual-MAD unit above the median fitted deviation. With

Dataset	nDCG@10	Δ	MAP@100	Δ
trec-covid	**71.37**	**+2.17**	**8.71**	**+0.20**
fiqa	**42.42**	**+0.73**	**36.34**	**+0.63**
hotpotqa	**73.29**	**+0.08**	**66.05**	**+0.02**
nq	**51.03**	**+0.05**	**44.02**	**+0.04**
nfcorpus	**36.53**	**+0.05**	**17.68**	**+0.17**
scifact	74.07	−0.12	69.78	−0.12
arguana	45.87	−0.42	32.80	−0.26
quora	87.17	−1.03	83.91	−1.19
Average	**60.22**	**+0.19**	**44.19**	−0.04

Table 13: BEIR evaluation with BGE-large and TRIX. Values are percentages ($\times 100$). Δ shows absolute change relative to the BGE-large baseline.

$\tau = 0.1$, each additional 0.1 active residual unit multiplies the affirmative reward by $\exp(-1)$. The full-query score, attenuated affirmative reward, and residual penalty use unit coefficients; no query-specific scoring weights are introduced.

F General Retrieval Evaluation

To assess whether TRIX affects general retrieval effectiveness, we compare the frozen BGE-large-en-v1.5 baseline with its TRIX-reranked results on eight BEIR benchmarks.

Table 13 reports nDCG@10 and MAP@100 for each dataset. TRIX maintains general retrieval effectiveness overall: five datasets improve in nDCG@10 (+0.05 to +2.17 points), while three show small decreases (-0.12 to -1.03 points). The average changes are +0.19 nDCG@10 and -0.04 MAP@100.

Across the eight datasets, TRIX preserves average general retrieval performance, improving nDCG@10 by 0.19 points while MAP@100 changes by only -0.04 points. NQ and HotpotQA show nDCG@10 changes of at most 0.08 points.